Mind It! v1.69 — QC Report (builds 121–129)

Date: 12 October 2026 · **Version:** 1.69 (unchanged) · **Android versionCode:** 121 · **Source:** git tag `checkpoint-build121`

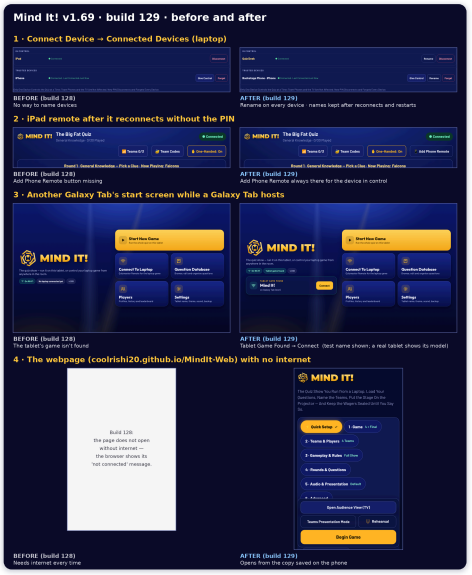
Where the testing happened: everything in this report was run on the build machine (Linux sandbox): the real Windows laptop app (Electron, the same app shell as `Mind It!.exe`) under a virtual screen, with the tablet and phone pages opened in Chromium at real sizes and touch emulation. **No physical Windows PC, Galaxy Tab A9+, phone or TV was used.** The live acceptance test is **PENDING USER LIVE TEST** (section J).

Overall readiness: Ready for your live acceptance test — not yet cleared for unrestricted live use. Every automated check, audit, stress test and the 60-minute soak passed in the sandbox, and the rollback was tested; what's missing is a real game on your real laptop, tablet, phones and TV (section J). Keep build 119 until that passes.

Build 129 addendum (16 October 2026)

Your choice: upgrades **1, 2, 4, 5, 6, 7**. Number 6 (smaller downloads) was dropped: the app's extra language files were already removed, so there was nothing left to trim — I told you at the time.

What changed - **1 · The Galaxy Tab keeps hosting in the background.** While the tablet hosts, Android is told a quiz is running: a notification *Mind It! is hosting a quiz*, and the tablet keeps its Wi-Fi and server awake if the screen goes off or another app is opened. Swiping Mind It! away stops it. Android 13+ asks once to allow notifications. - **2 · Other tablets find the tablet's game.** The hosting tablet announces its game on the Wi-Fi; another Galaxy Tab's start screen shows **Tablet Game Found** (and never lists its own game). Laptop games still say *Laptop Game Found*. - **4 · Rename devices** in Connected Devices (laptop and tablet): the device in control and every trusted device. The name is used on request cards, *Currently in control* and the waiting remote, and is kept through reconnects and restarts. An empty name goes back to the device's own name. - **5 · The webpage works offline.** The hosted webpage keeps a copy on the phone once opened; with no internet it opens from that copy (with its players and history). With internet it always loads the newest build first. - **7 · 📱 Add Phone Remote is always there** on the remote in control: the host now sends it the PIN, so the button no longer disappears after a reconnect without the PIN.



Before and after (real screenshots; the offline "before" is a description, because the browser shows only its own "not connected" message).

Check	Result
`fix129d` (laptop: Add Phone Remote after a reconnect without the PIN + its code decoded; Rename / Cancel / Enter / empty name; no redraw while typing; name in the request card and on the waiting iPhone; name saved in the trusted-devices file; kept after reconnect and laptop restart)	**21/21**
`fix129t` (the tablet's own server on the build machine: `kind: tablet` ; PIN sent to the controller; rename on the tablet, saved, kept after a server restart; another tablet's start screen shows *Tablet Game Found* + chip; the hosting tablet doesn't list itself; laptop games unchanged)	**16/16**
`fix129w` (the hosted webpage served as on GitHub Pages: offline helper installs; with the site switched off and the browser offline the page opens, with its saved data; a new tab opens offline; icon kept; other requests fail normally; back online a newer upload is used at once) — and a control run without the helper fails to open offline, as expected	**13/13**
Full regression	**103 suites, 2,316 / 2,316** (64 min). `fix127d` passed on retry: its wrong-PIN check reused a browser that has been trusted since build 128, so it sometimes connected without the PIN, which is correct. The check now uses a new device; re-run twice, **22/22** both times
Layout audits	host screens at 5 sizes + tablet home: **0 issues**; Audience View 234 screens at 9 sizes: **0 issues**
Android manifest	`foregroundServiceType=connectedDevice` on the hosting service, checked with `aapt dump xmltree` and with a second, independent reader (androguard); permissions FOREGROUND_SERVICE, FOREGROUND_SERVICE_CONNECTED_DEVICE, POST_NOTIFICATIONS; the Wi-Fi multicast permission Android 14 requires for this service type is declared

Check	Result
Windows EXE	69,016,865 bytes, SHA-256 `5d23dc2b…` ; tested page inside (md5 `ae10643d…`); `main.js` = build 129; parts re-join identically under Wine
Android APK	versionCode 129, SHA-256 `6b26e0b3…` , v2+v3, original key `5f4819e4…` ; tested page and new start screen inside
Mac (Apple Silicon)	arm64, ad-hoc signed, tested page + `main.js` inside; zip 100,269,100 bytes (95.6 MiB), SHA-256 `1e78e68d…` ; 5 parts re-join identically; **not run on a real Mac**

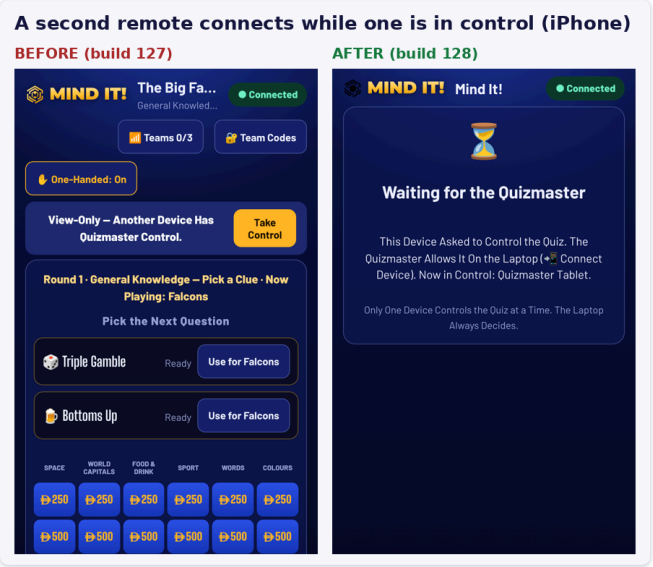
Limits of this testing — please read - 🟡 **The background hosting (1) and Tablet Game Found on a real network (2) were not tested at all.** They need a real Galaxy Tab running Android; the build machine has none. What was checked: the app is built as Android 14 and 15 require (manifest), and the code compiles. The tablet's server itself (rename, PIN) ran on the build machine. - **Rehearse at home:** start a quiz on the tablet, join with a phone, turn the tablet's screen off for a minute and check the phone still works; look for the notification; on a second tablet, check *Tablet Game Found*. - Offline webpage: tested in Chromium, not real iPhone Safari. An iPhone keeps the home-screen app's copy separately from Safari's. In Safari itself (not on the home screen), iOS can delete a site's saved data after 7 days without a visit, so use the home-screen icon.

Live acceptance: **PENDING USER LIVE TEST**.

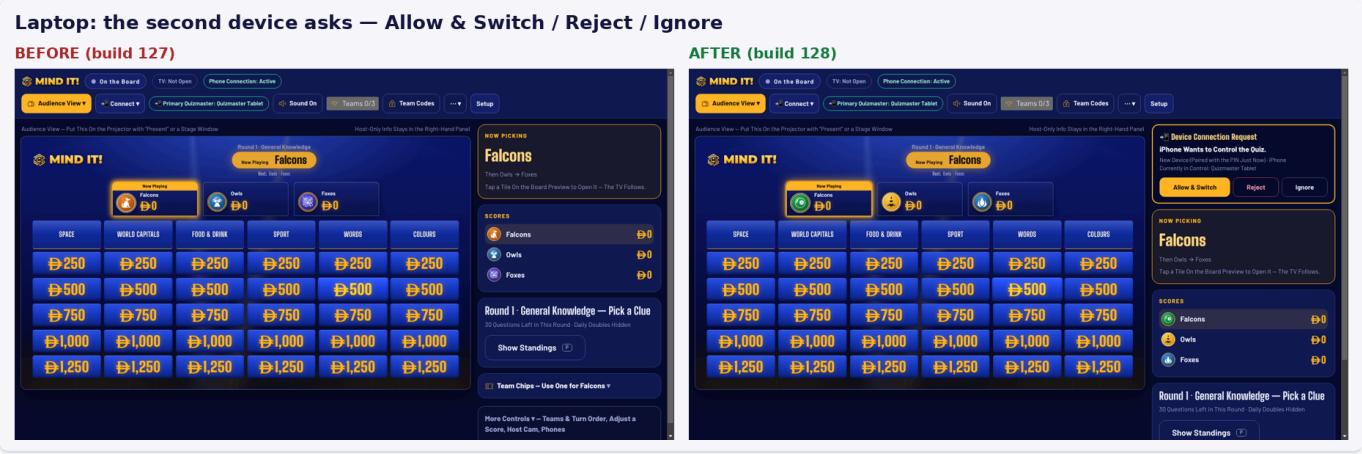
Build 128 addendum (16 October 2026)

Your two specs ("Trusted Device Reconnection & Host-Controlled Device Management" and "Multi-Device Remote Control Hierarchy") with your decisions: the 📱 Phone Remote counts as a remote; a phone joins the laptop directly; requests wait 2 minutes; and "i need the tablet to be the server".

What changed - **Trusted devices:** PIN once, then a device's own key; reconnects by itself after drops and restarts; Forget / New PIN revoke. - **One controller:** only one remote (tablet, iPad/iPhone or Phone Remote) controls the quiz; it keeps its place through drops and restarts; only it receives the game (answers stay private). - **Approval:** a second device waits; the host gets **Allow & Switch / Reject / Ignore** in the right-hand panel (never on the TV, not in Present mode); requests expire after 2 minutes. - **Connected Devices** panel: in control · waiting · trusted (last connected) · Give Control / Disconnect / Forget. - 📱 **Add Phone Remote** on a laptop-connected tablet sends a phone straight to the laptop. - **Galaxy Tab as the server:** the tablet app runs the same server as the laptop (rewritten in Java) when it hosts a quiz alone — team phones, chips, buzzers, Final answers, Phone Remote and remotes.



📸 *A second remote — before and after.*



📸 *The laptop when a second device asks — before and after.*

Connect Device → Connected Devices

BEFORE (build 127)



AFTER (build 128)



📸 *Connected Devices — before and after.*

Galaxy Tab hosting on its own: now the server (Connect Device)

BEFORE (build 127)



AFTER (build 128)



📸 *The Galaxy Tab hosting on its own — before and after.*

Your 14 + 15 test scenarios

Scenario	Where	Result
New device connects with the PIN; reconnects without it	`fix128d` 1–2	✅
Trusted device reconnects after the laptop app restarts (data kept)	`fix128e` 3	✅
Second device while one is active; Allow & Switch; Reject	`fix128d` 4–6	✅ the first device keeps working until you decide
A pending request expires without interrupting the quiz	`fix128e` 7	✅ (2 min; scores unchanged)
Manual switching from Connected Devices; Disconnect; Forget	`fix128d` 8	✅ a disconnected device asks before it gets control back
Active device loses Wi-Fi and reconnects	`fix128e` 9	✅ same question still open
The old device reconnects after a switch → asks	`fix128d` 10	✅
Several devices ask at once	`fix128d` 11	✅ exactly one in control
Revoked / wrong-PIN device; commands from a device that lost control	`fix128d` 12	✅ refused
Active device drops during a question and during a Daily Double wager	`fix128e` 13	✅ state preserved
Team phones and the Audience View unaffected	`fix128d` 14, `fix128e`	✅ (request never on the TV or in Present mode)
📱 Phone Remote counts as a controller	`fix128e`	✅ waits, Allow & Switch
Phone ↔ tablet: PIN, trusted reconnect, tablet hosts a whole quiz, phone controls it, switching, restart	`fix128t`	✅ on the build machine (JVM)
Tablet ↔ laptop, phone joins the laptop via the tablet's code	`fix128d` , `link`	✅

Check	Result
`fix128d` (laptop + iPad / iPhone / Android-sized pages)	**35/35**
`fix128e` (restart, Wi-Fi drop, expiry, Present mode, Phone Remote, drops mid-question / mid-wager)	**18/18**
`fix128t` (the tablet's own server — same Java source as the APK — on the build machine: phones join, DD wager incl. minimum, chips, buzzers, Final wager + answer, remotes, approval, restart, Check Connections, profiles)	**22/22**
Existing suites updated for the new rules	`link` (second tablet now waits; stale code with wrong PIN refused), `fix121` (Phone Remote approved first), `conn110`
Full regression	**100 suites, 2,269 / 2,269**, every suite first time (57 min)

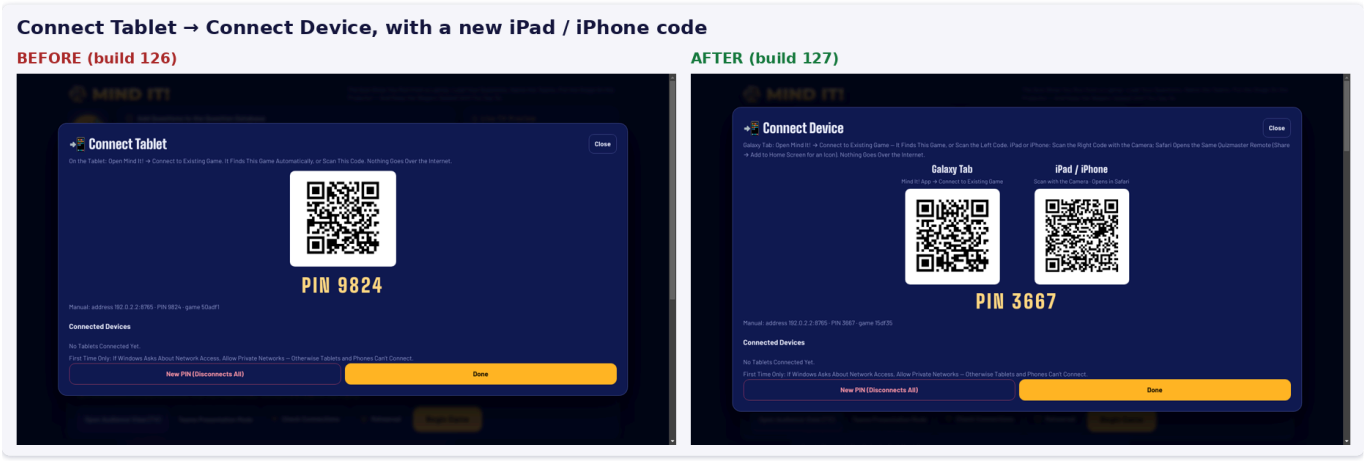
Check	Result
Layout audits	host screens at 5 sizes + tablet home: **0 issues**; Audience View 234 screens at 9 sizes: **0 issues**
Windows EXE	69,015,668 bytes, SHA-256 `1f38f139…` ; tested page inside (md5 `59efc466…`); `main.js` = build 128; parts re-join identically under Wine
Android APK	versionCode 128, SHA-256 `20ed7e1d…` , v2+v3, original key `5f4819e4…` ; tested page inside; the server classes inside
Mac (Apple Silicon)	arm64, ad-hoc signed, tested page + `main.js` inside; zip 100,268,686 bytes (95 MiB), SHA-256 `69f902de…` ; 5 parts re-join identically; **not run on a real Mac**

Limits of this testing — please read - 🟡 **The Galaxy Tab as the server has not run on a real Galaxy Tab.** The same Java code was tested on the build machine with the quiz page loaded as the app loads it and phones joining over the network. Android itself (the WebView, Wi-Fi, the app going to the background, power saving) was not part of it. - iPhones and iPads were simulated in Chromium, not real Safari. - "Encrypted communication" from your spec: connections stay on your local Wi-Fi over plain `http://` , as before. Pairing uses the PIN and a per-device key, and the laptop/tablet stores only a fingerprint (hash) of each key. Real encryption would need security certificates on every device, which can't be done offline without extra setup. - "Google Cast / Android TV audience views" from your spec: Mind It! doesn't have these, so nothing could be affected. The Audience View window works as before.

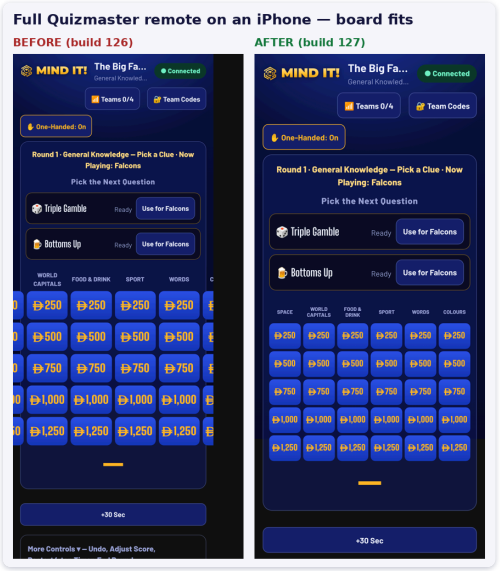
Live acceptance: **PENDING USER LIVE TEST** — especially a rehearsal with the Galaxy Tab as the server.

Build 127 addendum (16 October 2026)

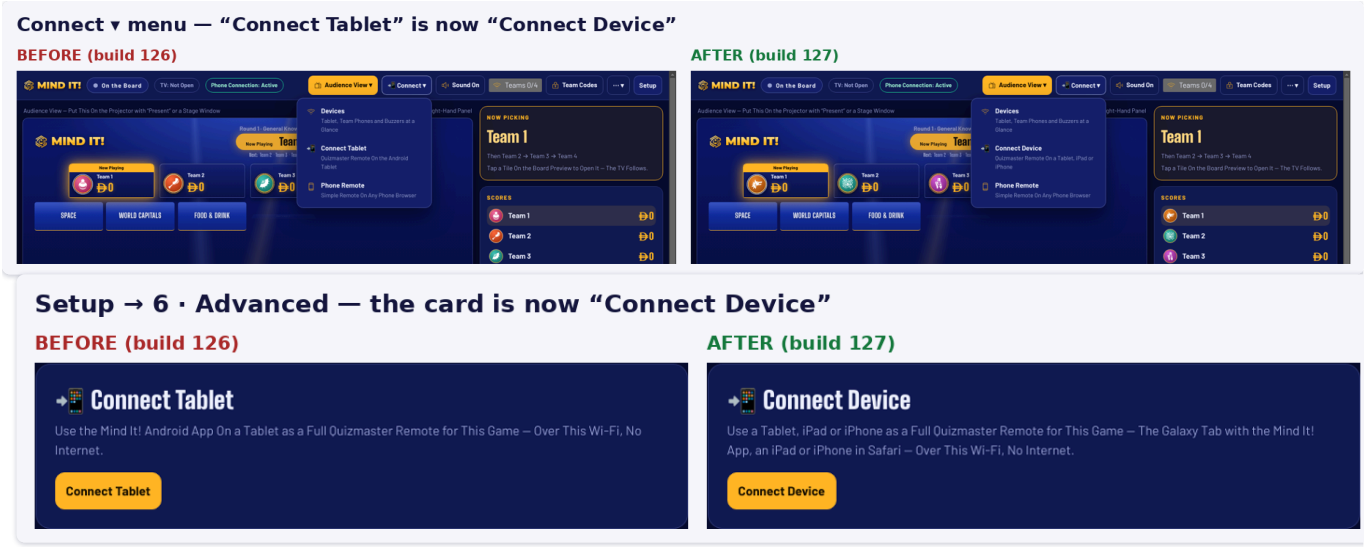
Changes (your requests: the iPad/iPhone remote — "do it"; then "change connect tablet into connect device"): - 📲 **Connect Tablet is now Connect Device** — in the Connect ▾ menu, Setup → 6 · Advanced, the Devices window (*Connect device, Show Device QR / PIN*) and the window itself. - The window shows **two codes**: **Galaxy Tab** (unchanged) and **iPad / iPhone** — a normal web address the Camera opens in Safari, giving the same full Quizmaster remote as the Galaxy Tab. The laptop app serves it at `/tablet` . - A home-screen icon for that remote keeps working after the laptop app restarts (it picks up the new game number; the PIN is still required). - On an iPhone the remote's board now fits the screen (all 6 columns). - **Bug found and fixed:** a restored player could count as still deleted if the deletion and the restore happened in the same millisecond (only reachable by automation — `fix125` failed once on its first try and passed on the retry). A restore is now always dated after its deletion.



📸 *Connect Tablet → Connect Device, with the new iPad / iPhone code — before and after.*



📸 *The full remote on an iPhone — before (board runs off the screen) and after.*



Connect ▾ menu and the Setup card — before and after.

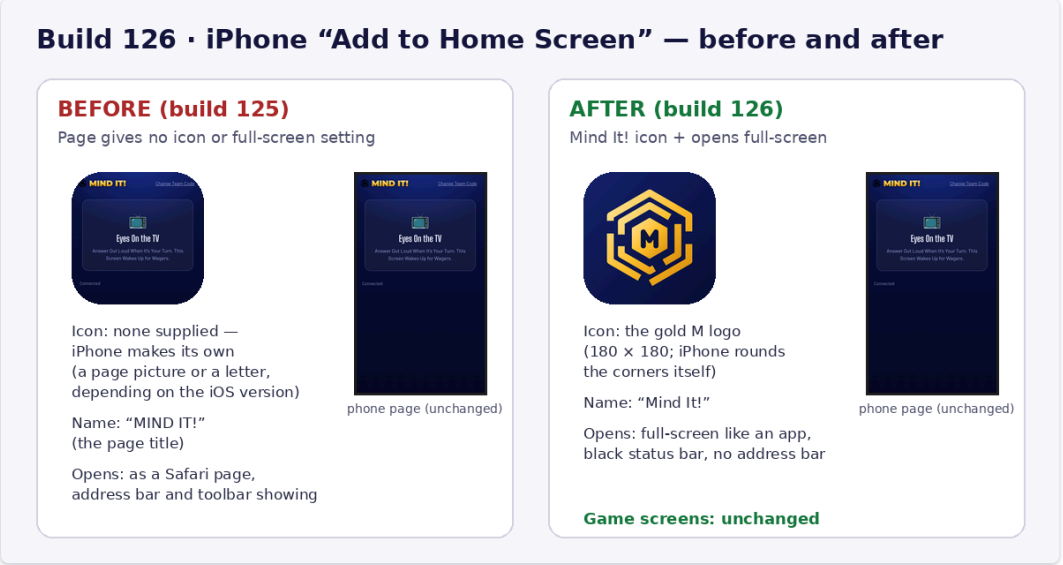
Check	Result
`fix127d` (laptop app + iPhone- and iPad-sized browsers): the window is Connect Device with two equal codes; the right code is the laptop's `/tablet` address with game number and PIN (both codes decoded from a screenshot and checked); the Galaxy Tab code unchanged; the dialog fits the laptop screen; Setup and the Connect ▾ menu say Connect Device; the laptop serves `/tablet` to other devices; iPhone: Connected, board fits (6 columns, nothing off the edges, every value in full), tapping a value opens the question on the laptop, Correct / Wrong appear; an old game number in the address still connects; a wrong PIN can't; iPad: connects, tablet layout unchanged; no errors	**22/22**
Full regression	**97 suites, 2,188 / 2,188** — 96 first time; `fix125` once on retry (the bug above), then fixed; after the fix `fix125` 3 × 27/27, `fix124` 19/19, `psync`, `libsync`, `fix125d`, `fix127d` 48/48
Layout audits	host screens at 5 sizes + tablet home at 3 sizes: **0 issues**; Audience View 234 screens at 9 sizes: **0 issues**
Windows EXE	69,009,786 bytes, SHA-256 `2470d7fa…` ; tested page inside (md5 `c9db9f0f…`), `main.js` with the `/tablet` route; parts re-join identically under Wine
Android APK	versionCode 127, SHA-256 `a547d00e…` , v2+v3, original key `5f4819e4…` , tested page inside
Mac (Apple Silicon)	arm64, ad-hoc signed, tested page and `main.js` inside; zip 100,261,994 bytes (95 MiB), SHA-256 `456f2c4e…` ; 5 parts re-join identically; **not run on a real Mac**
Webpage	on GitHub Pages: https://coolrishi20.github.io/MindIt-Web/ (updated to build 127)

Limits of this testing: iPhone and iPad were simulated with Chromium (size, touch, user agent), not real Safari. The GitHub-hosted webpage can't be a remote for the laptop — a secure `https://` page isn't allowed to talk to the laptop's local `http://` address; use the **iPad / iPhone** code instead.

Live acceptance: **PENDING USER LIVE TEST**.

Build 126 addendum (13 October 2026)

Change (your request: "ok do both, include the webpage as a download option in github"): 📲 Mind It! on the home screen — on an iPhone or iPad, Safari's **Share → Add to Home Screen** now gives a **Mind It!** icon (the gold M logo) that opens the team-phone page or the 📱 Phone Remote full-screen. The page's `<head>` only: no game screen, rule or saved-data change. The webpage version `mindit.html` is now also on your GitHub download page.



📸 *Home-screen icon and launch — before (build 125) and after (build 126).*

Check	Result
`fix126` (the page itself): full-screen setting, name "Mind It!", browser tab title unchanged; 180×180 and 192×192 icons solid (no see-through corners); page starts normally at laptop and iPhone size, no errors	**8/8**
`fix126d` (laptop app + an iPhone-sized phone): the laptop's `/play` and `/remote` pages carry the full-screen setting, name and icon; the icon is the gold M logo; with the "opened from the home screen" flag set, the phone joins its team and shows "Connected"; nothing sticks out sideways; the Phone Remote still connects with its PIN; no errors	**13/13**
Webpage as a downloaded file	opened from disk in Chromium: starts, keeps a saved game and a new player after closing and reopening, Audience View opens; no errors
Full regression	**96 suites, 2,169 / 2,169**, every suite first time (53 min)
Layout audits	host screens at 5 sizes + tablet home at 3 sizes: **0 issues**; Audience View 234 screens at 9 sizes: **0 issues**
Windows EXE	69,008,995 bytes, SHA-256 `6cd8baec…` ; tested page inside (md5 `d9d70e55…`), `main.js` unchanged; parts re-join identically under Wine
Android APK	versionCode 126, SHA-256 `c18ac5de…` , v2+v3, original key `5f4819e4…` , tested page inside
Mac (Apple Silicon)	arm64, ad-hoc signed, tested page inside; zip 100,260,337 bytes (95 MiB — under GitHub's 100 MiB limit), SHA-256 `c04447c1…` ; 5 parts re-join identically; **not run on a real Mac**
Web link	republished (alternate builds: 116, 118, 120, 122, 124, **126**); the published page carries the new head tags and the same app code

Limits of this testing: the phone checks used Chromium pretending to be an iPhone (size, touch, iPhone user agent). Real Safari wasn't available here, so how iOS shows the icon and opens it full-screen is **not verified on a real iPhone or iPad**. Android Chrome opens a plain `http://` address from the home screen as a normal tab (no full-screen).

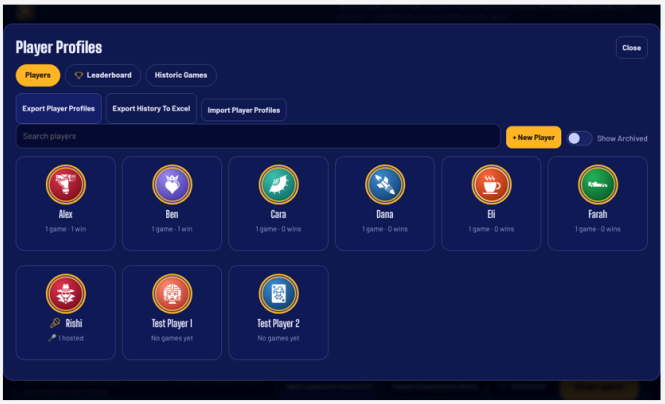
Live acceptance: **PENDING USER LIVE TEST**.

Build 125 addendum (13 October 2026)

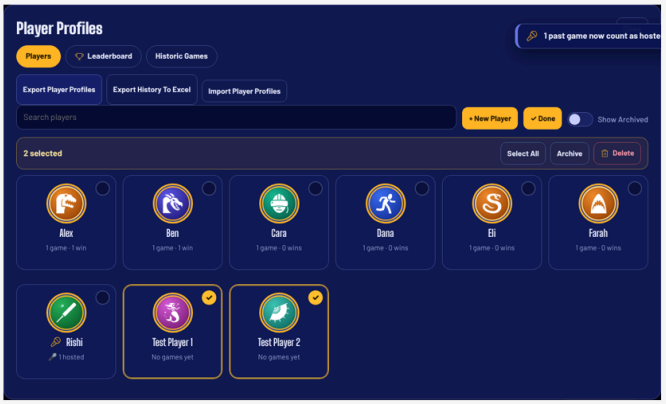
Changes (your request, all five): 🗑 Recently Deleted (restore a deleted player for 30 days) · 🛟 Safety backups (automatic, last 10) · ☑ Select several players · the tablet keeps what you type · 🖼 Results Picture for any Historic Game.

BEFORE — build 124

1+3 · Player Profiles: “Select” (tick several → Archive / Delete) · “Recently Deleted” appears after a deletion



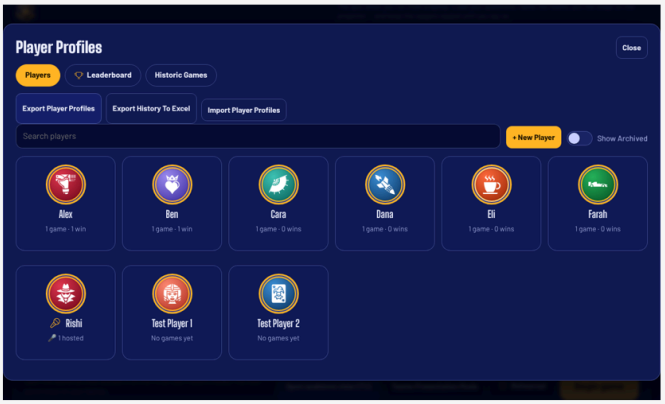
AFTER — build 125



📸 *Player Profiles — before (build 124) and after (build 125, Select mode with two players ticked).*

BEFORE — build 124 (no way back)

1 · Recently Deleted: Restore for 30 days



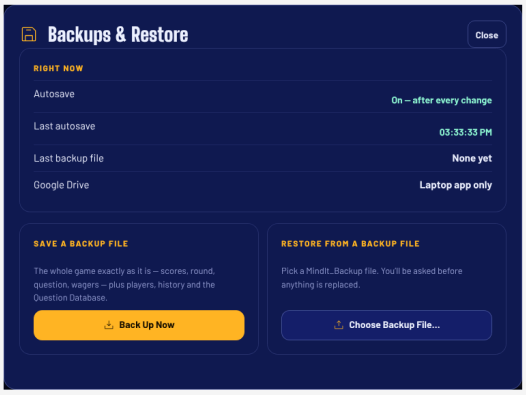
AFTER — build 125



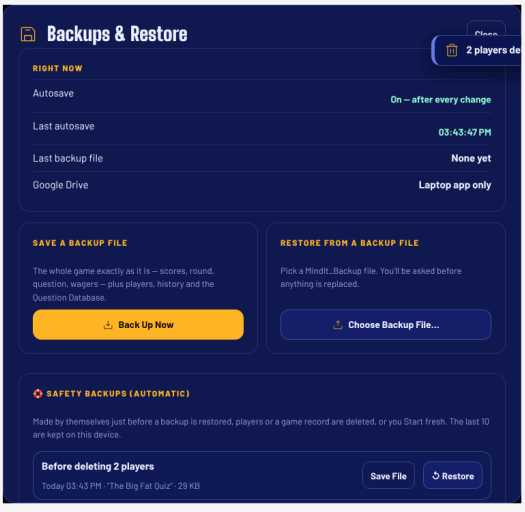
📸 *Recently Deleted (new).*

BEFORE — build 124

2 · Backups & Restore: Safety backups (automatic, last 10)



AFTER — build 125



📸 *Backups & Restore — before and after (🛟 Safety backups card).*

BEFORE — build 124

5 · A Historic Game: “Results Picture” joins the bottom row



AFTER — build 125



📸 *A Historic Game's bottom row — before and after.*

Check	Result
`fix125` (browser): Select / Select All / untick / Delete several with one question / Archive several; Recently Deleted list, Restore, picture and nickname back, 30-day expiry keeps the name, a sync can't re-delete a restored player; safety backups before deleting players, a game record and Start fresh, only 10 kept, listed in Backups & Restore, Save File gives a normal backup, Restore asks first; Results Picture saves a PNG for a past game	**27/27**

Check	Result
`fix125d` (laptop app + tablet): Adjust Score — amount, reason, chosen team and the cursor survive the laptop scoring meanwhile; Cancel starts empty next time; Apply works	**9/9**
Full regression	**94 suites, 2,127 / 2,127**, every suite first time
Layout audits	host screens at 5 sizes: **0 issues**; Audience View 234 screens at 9 sizes: **0 issues**
Windows EXE	68,963,836 bytes, SHA-256 `e995b25f…` ; tested page inside; parts re-join identically under Wine
Android APK	versionCode 125, SHA-256 `e39b61b4…` , v2+v3, original key
Mac (Apple Silicon)	arm64, ad-hoc signed, tested page inside; 5 parts re-join to `5fcb4496…` ; **not run on a real Mac**

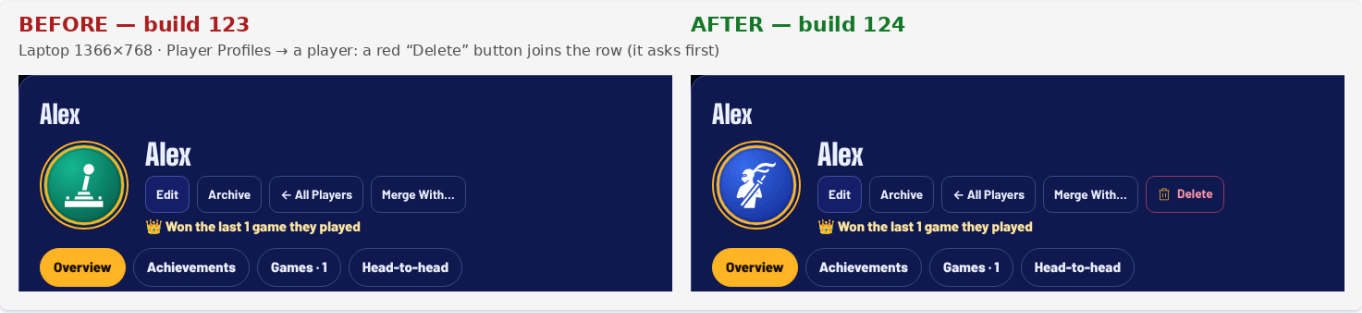
Note: safety backups are stored inside the app on each device (laptop app, or the tablet when it runs a game alone) — not on the linked tablet remote and not in a file you can see; **Back Up Now** files are still the backup to keep elsewhere.

Live acceptance: **PENDING USER LIVE TEST**.

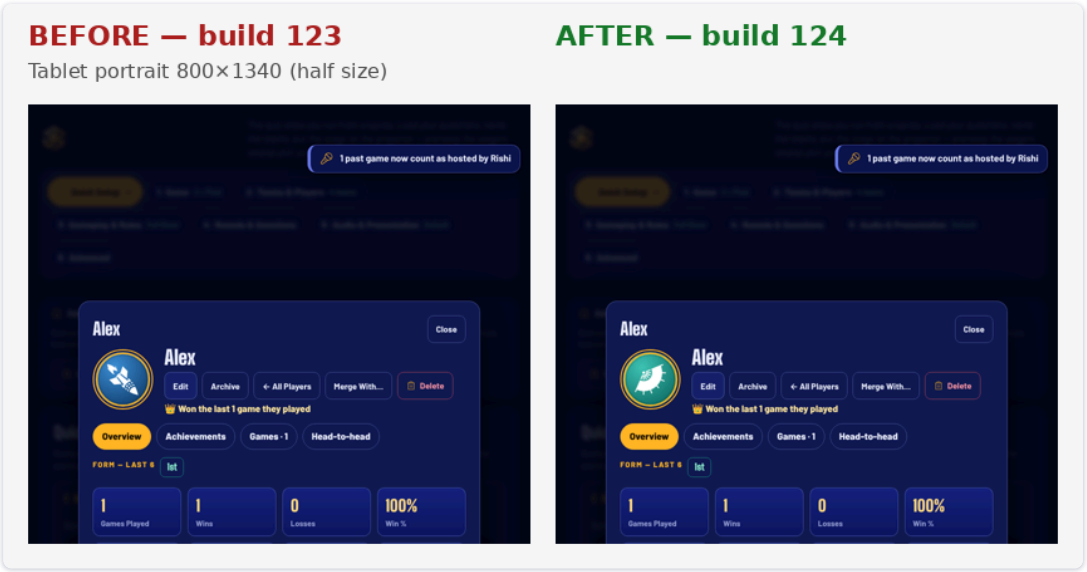
Build 124 addendum (13 October 2026)

Changes (your requests): 🗑 **delete a player** from Player Profiles; the Mac app is now **Apple Silicon only**.

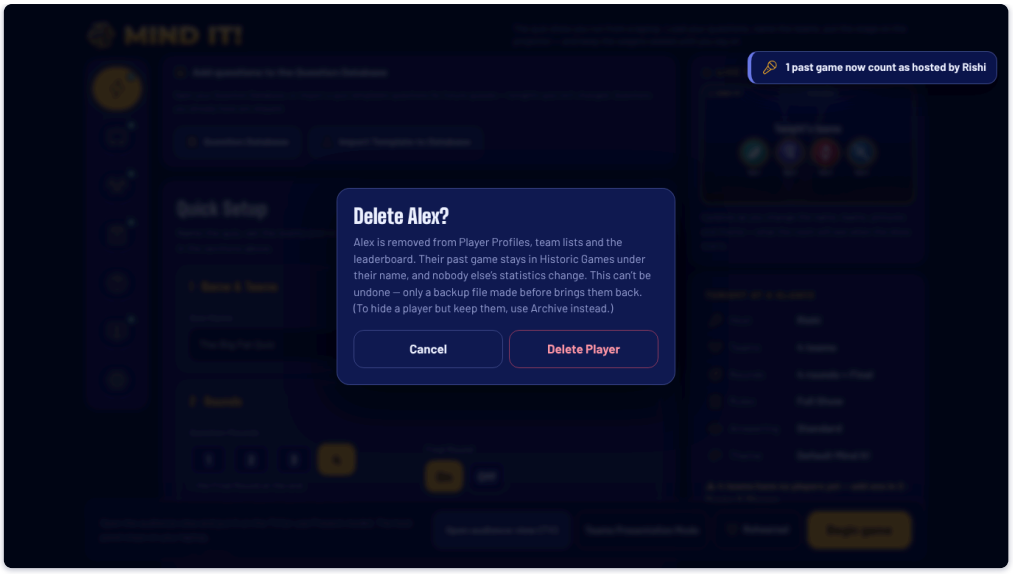
Layout — before / after (the only layout change is one more button on a player's page; nothing else moved):



📸 *Laptop 1366×768 — a player's page, build 123 (left) and build 124 (right).*



📸 *Tablet portrait 800×1340 — the same row fits on one line.*



The question before deleting.

Check	Result
New suite `fix124` : button on the same row as Edit/Archive/All Players/Merge; asks first; Cancel keeps; Delete removes from Player Profiles, the list and team lists; on Setup taken off their team; Quizmaster cleared; refused during a game; Historic Games keeps the name; other players' statistics unchanged; survives a reload; sync never brings the player back and passes deletions on; Returning Champion ignores deleted winners; restoring an older backup brings the player back	**19/19**
Full regression	**92 suites, 2,097 / 2,097**, every suite first time
Layout audits	laptop/tablet/phone host screens (incl. the player page) at 5 sizes: **0 issues**; Audience View 234 screens at 9 sizes (FHD, 4K, 16:10, 4:3, portrait): **0 issues**
Windows EXE	68,959,783 bytes, SHA-256 `bd485b3b…` ; tested page inside; parts re-join identically under Wine
Android APK	versionCode 124, SHA-256 `a0ca8c0a…` , v2+v3, original key `5f4819e4…`
Mac (Apple Silicon)	arm64, ad-hoc signed, tested page inside; zip 100,212,557 B (symlinks kept), 5 parts re-join to `fbdf9b69…` ; **not run on a real Mac**
Web link	republished (version 166), live page verified

Live acceptance: **PENDING USER LIVE TEST**.

Build 123 addendum (13 October 2026)

Changes (your decisions): restoring a backup or loading a game file **always asks first** (Restore Backup / Load File / Cancel), even with no game running — K4 settled; team photos stay in the save (K3, "leave it"); a **Mac version** of the laptop app (your request).

Check	Result
New suite `fix123` : asks with no game running; Cancel leaves the quiz untouched; Restore Backup restores game and players; with a game in progress only the usual "A game is paused…" question; game files ask "Load this game file?"; player-profile files merge without asking; Backups & Restore → Choose Backup File… asks; the in-app note is now true	**13/13**
Full regression	**91 suites, 2,081 / 2,081.** `qlib` and `crop` restore a backup as part of their checks; they first failed because the new question appeared, were updated to press Restore Backup, then passed (26/26, 27/27).
Windows EXE	68,958,930 bytes, SHA-256 `463b707e…` ; page md5 `feeb27bd7019…` inside; parts re-join identically under Wine
Android APK	versionCode 123, SHA-256 `b937f559…` , v2+v3, original key `5f4819e4…` ; page md5 `feeb27bd7019…`
Mac app — Apple Silicon	arm64, bundle `com.mindit.quiz` 1.69.0, macOS 11+, camera and local-network permission texts present, ad-hoc signed (app and frameworks), tested page and `main.js` inside; zip 100,211,536 B with the framework symlinks kept; 5 parts re-join to the same SHA-256 (`51b4421c…`)
Mac app — Intel	x86_64, same checks; zip 106,274,619 B (`18767abd…`)

Check	Result
Mac app on a real Mac	**Not run** — no Mac here. It isn't notarized by Apple, so macOS asks once (Open Anyway).
Guides	Complete Guide 4.1b (Mac install), 22.3/22.5 (restore asks), Quick Start Mac line; Cheat Sheet and Player's Guide still exactly 2 pages

Live acceptance (Windows, Galaxy Tab, phones, TV — and now the Mac): **PENDING USER LIVE TEST**.

Build 122 addendum (13 October 2026)

Change: your Returning Champion rule — the crown goes only to a team that has **every player of last game's winning team**; a one-player winner crowns the team that player is on; split winners → no crown (`app111.js`). This settles K1.

Check	Result
New suite `champ122` (split winners, full team, full team + new player, 1 of 4 / 2 of 4, one-player winner, deleted player, teams without players, joint winners, win streak and where it stops, rule switched off)	**15/15**
Existing `champ` (crown, TV intro, Title Defence, Dethroned, next champion)	9/9
Full regression	**90 suites, 2,061 / 2,061** (`desk` passed on retry: its first attempt couldn't connect to the freshly started laptop app)
Windows EXE	68,959,626 bytes, SHA-256 `2158c7c4…` ; page md5 `e2d075a926a6…` inside; parts re-join identically under Wine
Android APK	versionCode 122, SHA-256 `a4481b76…` , v2+v3, original key `5f4819e4…` ; page md5 `e2d075a926a6…`
Web link	republished (version 165); live page verified

Everything else in this report (build 121) still applies. Live acceptance: **PENDING USER LIVE TEST**.

A. Previous task completion

Interrupted task	Build 120 — "Interactive team swapping & Returning Champion logic" (your attachment 1): click-to-select / swap / + move / undo / reset / confirm in the Random Team Generator, with persistence, tests and layout testing; plus a review (not a change) of the Returning Champion logic.
Done before your QC message arrived	The team-swapping code (`app109.js`), its suites and the desktop/tablet layout check were finished; the Returning Champion review was written up.
Remaining when the QC message arrived	The build-120 EXE/APK, the full regression run, the guide update and the deliverables A–F.
Resumed and finished	Build 120 was built, verified and checkpointed (EXE `e93ffe65…` , APK `d65fd147…` vc120, web link v164); full run 2,019/2,019 (two suites passed on retry — the cause was found and fixed in this QC, see C2); the guide's §7.4 got the new team-swapping text and screenshot.
Verification of the build-120 requirements	`rtg120` **52/52** (mouse 1366×768 and portrait-tablet touch 800×1280): select, swap, same-team re-select, cancel, + move (refused when it would empty a team), undo (swaps and moves), reset (asks first), confirm saves the final teams, scroll position kept, 46 px+ touch targets. `fix120` 13/13. In build 121 both pass again (section B).
Returning Champion	Reviewed only — **not changed**, because you asked for approval first and no new rule has been approved. The current rule is now described exactly in the guide (§12.5). See K1.

B. QC results

Full automated regression, build 121 (final page): 89 suites (55 browser + 34 laptop-app), 2,055 / 2,055 checks passed, every suite first time (no retries), 49 minutes. (An earlier full run of build 121, before fix C7 was added, gave 2,025/2,025 in 88 suites + `fix121` 32/32. `qc2` plays random games, so its check count varies from run to run: 276–291.)

Area	Suites (examples)	Result
Game creation, Setup, rounds, Question Database	`qc2` , `zones` , `qlib` , `ext` , `l3` , `l4` , `fix120`	647/647 (7 suites)
Teams, players, Random Team Generator	`rtg` , `rtg120` , `g1` , `g3`	89/89 (4 suites)
Gameplay rules (scoring, Daily Double, Final, chips, buzzer, ties)	`rules35` , `tg` , `tg2` , `ud` , `bu` , `bz` , `bz3` , `r9` , `fix112` – `fix115`	250/250 (12 suites)
Historic Games, statistics, achievements, champion	`champ` , `g2` , `g6` , `g7` , `rh` , `v169`	74/74 (6 suites)

Area	Suites (examples)	Result
Phones, tablet, link, End Game	`link`, `pin`, `psync`, `hostreq`, `hostreq2`, `conn110`, `endgame110`, `fix116`, `fix117`, `fix119`, `fix121`	238/238 (11 suites)
Complete games end to end	`fullmatch` (whole match from the tablet), `onehand`, `desk`	64/64 (3 suites)
Audience View layout	`avaudit` — 234 screens × automatic checks at 9 sizes	**235/235**, 0 layout issues
Laptop host screens	`vaudit` — 18 screens × 5 sizes (90 screenshots)	**0 issues**
Live-game stress and soak	`stress110` phases	**24/24 + soak 2/2:** idle 2 × 10 min with the Quizmaster window hidden (117 samples each, TV + tablet + 10 phones: no false disconnects, no full-screen cover); network drop/return 6/6; 10 phones at once 4/4; phone lock 60 s 2/2; rapid/duplicate tablet taps 4/4; Check Connections 6/6; **60-minute real-time quiz:** 60 questions, 346 tablet actions, Quizmaster window hidden/shown 5×, an intermission — **0 false disconnects, 0 script errors**, laptop server memory 213 → 222 MB, page heap 12 → 13 MB
Rollback (build 121 data → build 119 app)	rollback test, section I	11/11

Skipped / not possible here: real hardware (Windows PC, Galaxy Tab A9+, phones, TV, real Wi-Fi, real sound); the Android rollback; Google Drive backup (needs internet and your script).

End Game (asked specifically): `endgame110` 19/19 — End Game is refused while the game is still being played; the game is recorded **once**; the button locks while saving (no double tap); a repeated End Game after a dropped connection is acknowledged without saving twice; an End Game for an old game is refused; rehearsal End Game records nothing; standalone tablet saves once.

C. Bugs found and fixed (build 121, `app110.js`)

Every item was reproduced or measured first, then fixed, then re-measured. No gameplay rule, layout or saved-data format was changed.

#	Defect (confirmed)	Effect on a quiz night	Fix	Evidence
C1	**Connection Health: phones flickered Responsive ↔ Unstable** although they checked in every 1.5 s. The laptop added the time since its own 3-second health refresh to each phone's age (1.5 s + 3 s > the 4 s limit).	False "Unstable" warnings; the tablet's 📶 chip changed colour for nothing.	Only time beyond the normal refresh counts. A phone that really stops still turns amber (after ~7 s) and red/stale (~15 s).	80/80 samples "ok" over 20 s for each phone (before: flapping every second); dropped phone → Unstable at 7 s, Stale at 19 s.
C2	**Quizmaster tablet redrew its whole screen ~once a second.** The laptop re-sent the whole game because round-trip times inside Connection Health changed at every check-in.	Typed Adjust Score numbers could vanish; a tap could land on a button that was being replaced (this is why `hostreq2` and `fullmatch` needed a retry in the build-120 run).	The tablet's copy of Connection Health uses steady wording (round-trip bands, "just now"); the laptop keeps exact numbers.	Tablet state messages and full redraws: **1.07/s → 0/s** at idle; data sent to the tablet **~14.5 KB/s → 0**; `hostreq2` and `fullmatch` now pass first time.
C3	**Phone Buzzer Mode: BUZZ IN was rebuilt ~6.5 times a second**, so a tap whose press and release straddled a rebuild never arrived.	Lost buzzes — a fairness problem in a buzzer race.	The button is rebuilt only when something on it changes.	30 ordinary taps (120 ms): **14/30 → 30/30** reached the button; rebuilds **6.52/s → 0**. A real buzz still registers; the fast check pauses while the phone is locked.
C4	Phone screen rebuilt at **every check-in during a question** (the seconds left were part of the "has anything changed?" test).	Same lost-tap risk on chip/confirm buttons; wasted battery.	Seconds left no longer count as a change; the countdown runs on the phone and still follows +time / stop from the laptop.	Rebuilds **0.67/s → 0** in 6 s; +15 s reached the phone within 0.1 s of the laptop; stop cleared the phone timer.
C5	**Laptop Phones card (and wager and live-setup cards) rebuilt every 500 ms.**	"Join link & QR code" snapped shut; the network-address list closed while open; clicks during a rebuild were lost.	A card is replaced only when its content changes; open sections stay open; nothing is replaced while you type or choose in it.	Phones card rebuilt **2.0/s → 0/s**; a phone joining still updates it at once.
C6	📱 **Phone remote: every button press started another background check loop**; the page was redrawn every second while the timer ran.	After a few presses the phone hammered the laptop; presses could be lost.	One check loop; a press just makes the next check come sooner; the seconds update in place.	6 presses → 10 checks in 7 s (one loop); no redraw just for the seconds.
C7	**An unfinished game was lost if Mind It! was opened and closed again without choosing** Resume saved game / Start fresh. At start-up the app shows a fresh Setup while offering the saved game; closing saved that fresh Setup over it. **Also in build 119.**	A crash, reopen, close (e.g. to fix the TV), reopen → the game was gone (only a backup file could bring it back).	The unfinished game is kept aside until you choose; it is offered at every start. Save format unchanged; the extra copy (`mindit.resume.pending`) is ignored by older builds.	Restart test on the real laptop app: build 119 — offered once, then **never**; build 121 — offered at the 1st, 2nd and 3rd restart; Resume → right scores; Start fresh → no more offer.

Regression suite added: `fix121` (39 checks) covers C1–C7.

Found, not changed (reported): see K.

D. Connectivity results

Device	What was checked (simulated)	Result
Windows host (laptop app server)	Health endpoint, port fallback, saved-game and settings copies across a port change (`fix115` , `fix116`), Check Connections	Passed (`fix115` 22/22, `fix116` 26/26, stress `health` 6/6: host, tablet, TV and 8/8 teams green in 3.6 s)
Quizmaster tablet	Pairing/PIN, control hand-over, commands with acknowledgements, reconnect after Wi-Fi loss, End Game retry, Connection Health sheet (`link` , `pin` , `hostreq` , `hostreq2` , `conn110` , `endgame110` , `fix121`)	Passed (`link` , `pin` , `hostreq` 28/28, `hostreq2` 29/29 first time, `conn110` 35/35, `endgame110` 19/19, stress `net` : offline 20 s → small status line only, back in 0.3 s with the current state)
Audience View	No full-screen "Reconnecting" cover; heartbeat; the Quizmaster window hidden/minimised (build-110 fix) — `conn110` , stress `idle` / `soak`	Passed — no full-screen cover in 2 × 10 min idle and the 60-min soak with the Quizmaster window hidden half the time
Phones	Join, team codes, wagers, chips, buzzer, lock/unlock, network drop/return, Connection Health status (`psync` , `bz` , `conn110` , stress `lock` / `net` , `fix121`)	Passed (`psync` , `bz` 3 phones, stress `load` 10 phones at once, `lock` 60 s locked → "asleep", back on the same team with no re-pairing; `fix121` C1/C3/C4)
False "Reconnecting" (your live-game report)	The root cause found in build 110 (the Quizmaster window throttled when covered) stays fixed; build 121 removes the remaining false **Unstable** flicker (C1).	0 false disconnects in about 1 h 25 min of stress and soak sampling (2 × 10 min idle, the 60-min quiz and the short phases); Connection Health stays "Connected" for phones checking in normally (80/80 samples)

E. Performance results

Measured on the build machine with the laptop app, one phone and the tablet, 30 s per phase (`perfmeasure.py`), before and after build 121:

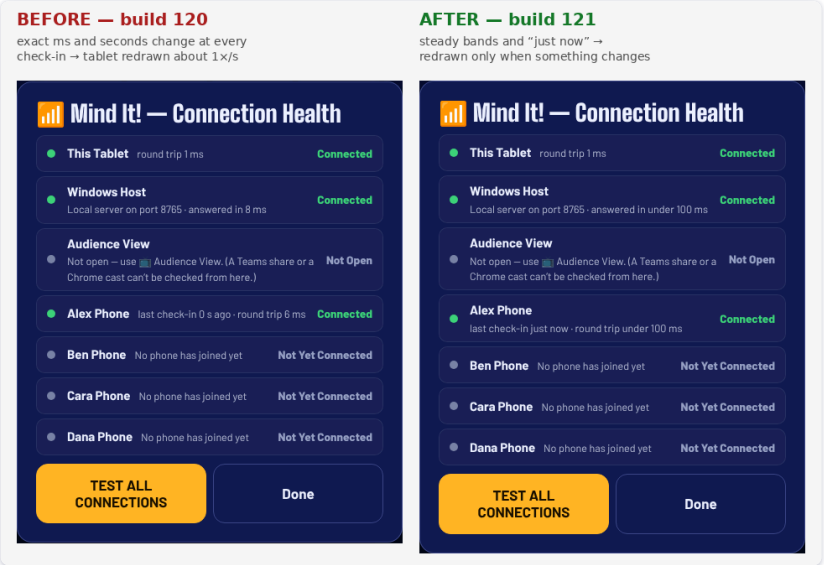
Phase	Measure	Before (build 120)	After (build 121)
Board idle	Tablet full redraws / s	1.07	0
Board idle	Game updates sent to the tablet / s · bytes / s	1.07 · 14,540	0 · 0
Board idle	Laptop Phones card rebuilt / s	2.0	0
Question, timer running	Tablet full redraws / s	1.07	0
Question, timer running	Phone screen rebuilds / s	0.67	0
Phone Buzzer Mode	BUZZ IN rebuilt / s · taps that arrived	6.52 · 14/30	0 · 30/30
Phone remote	Checks / s after 6 presses	grew with every press	~1.4 (one loop)

Requests per second are unchanged (phones check in every 1.5 s, 350 ms in Buzzer Mode); what changed is how often screens are thrown away and rebuilt.

Autosave: with four 300 KB team photos a save is ~1.2 MB and takes ~23 ms here; saves happen 0.7 s after a change (not on a timer — 0 saves in 20 s idle). Not changed (K3).

F. Visual QC

No layout was changed in build 121 (so there is no layout before/after). The only visible text change is in the **tablet's** Connection Health sheet — before/after:



Tablet → Connection Health, build 120 (left) and build 121 (right), same devices.

Screen	Sizes tested	Result
Audience View (TV)	1920×1080 (FHD), 3840×2160 (4K), 2560×1600 and 1920×1200 (16:10), 1280×800, 1366×768, 1280×720, 1024×768 (4:3), 800×1280 (portrait) — 26 key screens each	234 screens, **0 issues** (text off screen, cut off, overlapping; board tiles equal and centred; smallest text)

Screen	Sizes tested	Result
Laptop host	1920×1080, 1366×768, 1340×800, 800×1340, 390×844 — 18 screens	**0 issues**
Quizmaster tablet (Galaxy Tab A9+ size, 1340×800 / 800×1340)	`fix119`, `rtg120`, `endgame110`, `butab` layout checks	passed (in the full run)
Phones (390×844 @2×)	`fix119`, `fix116`, `psync`	passed (in the full run)

Screenshots are from the sandbox; real-screen colour, brightness and viewing distance still need your rehearsal (Rehearsal Checklist).

G. Documentation update

Document	Updated for build 121	Check
Complete Idiot's Guide (PDF + Markdown)	"New in build 121"; §9.2 tablet Connection Health wording; §12.5 exact Returning Champion rule (current behaviour, 🔵 review pending); §22.3 restore wording corrected (asks only when a game is in progress; players/history merged); crash-recovery note (the offer now stays)	138 pages, rebuilt and text-checked
Quick Start	build stamp	3 pages
Quizmaster Cheat Sheet	build stamp	**exactly 2 A4 pages**, no overflow
Player's Guide	unchanged content (nothing changed for players)	**exactly 2 A4 pages**, no overflow
Rehearsal Checklist	"stay Connected for a minute, no flicker"; reopen twice without choosing; restore asks mid-game	2 pages
Project docs (status, changelog, rules, architecture, CLAUDE.md)	B31–B40, build 121 row, Build-121 module notes, spare-copy key	—
Documentation audit	new section for builds 116–121	—

Every build-121 change is described in the guide's "New in build 121" and in the changelog. Code vs documents: where they disagreed, the code was taken as the truth and the documents corrected (restore wording, Returning Champion detail); one in-app sentence is still too broad (K4).

H. Build verification

	Windows	Android
File	`Mind It!.exe` (portable, x64) — also delivered as `MindIt_v1.69.exe` (same bytes) and as 3 parts + `Join Mind It - build 121.bat`	`Mind It! for Android (Galaxy Tab).apk` — also as `MindIt_v1.69.apk` (same bytes)
Size	68,959,878 bytes; parts 3 × 22,986,626	1,574,609 bytes
SHA-256	`d8d14ca1a12187d44343eb52aa786c13a49aed15c71ee0ff6f0ac3c7e4a2f12e`	`67926597a774e18a8f84e6e43fe63c1ca2e35a1ec0a90a9348427ce68b274b62`
Version	1.69 (package 1.69.0)	`com.mindit.app`, versionName **1.69**, versionCode **121**
App page inside	`app.asar` → `app/mindit.html` md5 `a146a07d0bbc…` = the tested page; `main.js` identical to build 119	`assets/mindit.html` md5 `a146a07d0bbc…` = the tested page
Signing	not code-signed (as before — SmartScreen "More info → Run anyway")	v2 + v3 verified; one signer, certificate SHA-256 `5f4819e4b15e…` = your original key
Checks run	Built with electron-builder (exit 0); app.asar extracted and compared; parts re-joined with the `.bat` under Wine → identical SHA-256	`aapt dump badging`, `apksigner verify -v --print-certs`, page md5
Not run	The EXE on a real Windows PC (the same app shell and page ran all laptop tests here under Linux)	Install/run on a real tablet (no device or emulator here)

Source: `MindIt_Source_v1.69.zip` (7.0 MB, 438 files, extracts cleanly). Rebuilding the app page from the extracted zip gives a file identical to the shipped page except 3 characters — the timestamp inside the compressed HEIC helper (its unpacked content is identical). Git commit `c926b63`, tag `checkpoint-build121`.

I. Rollback verification

- **Preserved:** `MIND_IT_v1.69_PREVIOUS_WORKING_RELEASE.zip` (build 119 as delivered — not rebuilt): 86,577,817 bytes, SHA-256 `a745c41b…`; 390 files, all checksums verified after unzipping; the EXE parts re-join to `49517ba1…`. Kept separate from the build-121 files and not overwritten. Git tags `checkpoint-build119`, `checkpoint-build120`, `checkpoint-build121`.
- **Rollback test (laptop app, sandbox):** build 121 wrote 5 players, a finished game, an unfinished game and a backup file; the app was closed, the build-119 page put in place and the app started on the **same profile**: players and Historic Games kept, the finished game's scores intact, the unfinished game offered and resumed with the right scores, and the build-121 backup file restored through ⋯ → Backups & Restore — **11/11**. (Run twice: with the build-121 page before and after fix C7.)
- **Android:** not tested (no tablet). Android refuses to install an older versionCode over a newer one; uninstalling deletes the tablet's Mind It! data. The guide gives three options, data-safe first.

J. Live acceptance status

PENDING USER LIVE TEST. Nothing here was run on your Windows laptop, Galaxy Tab A9+, phones or TV. Please play a 60–120-minute game with the real devices (the Rehearsal Checklist covers it) and watch for: false disconnects, full-screen Audience View interruptions, lost scores, duplicate actions, wrong game records, crashes, devices that don't reconnect. Keep the build-119 rollback until it passes.

K. Outstanding issues and risks

#	Issue	Severity	Status
~~K1~~	**Returning Champion (settled in build 122 — see the addendum):** a team with **any one** of last game's winners qualifies, and only the **first** qualifying team (team order) gets the crown — so a team with 1 of 4 winners can beat a team with 2 of 4. The proposed majority rule has open edge cases (tied winners, small rosters, deleted players, teams without players, the "N wins in a row" count).	—	**Done in build 122**
K2	Real-hardware behaviour (Wi-Fi quality, Android power saving, TV scaling, sound levels)	—	PENDING USER LIVE TEST
K3	Autosave includes team photos (~1.2 MB / ~23 ms per save with 4 photos).	Low	**Kept as is — your decision (13 Oct)**
~~K4~~	In-app Backups & Restore note said "You'll be asked before anything is replaced" but it asked only when a game was in progress.	—	**Fixed in build 123: it always asks now**
K5	The Android shell doesn't pause the page when the tablet app goes to the background (keeps the link alive — intended).	Info	Not changed
K6	The tablet still redraws fully when the game really changes (a score, a new question), so typing in Adjust Score while another device scores can still be interrupted.	Low	Not changed (rare; the once-a-second redraw is gone)
K7	Test scripts and `build.py` use the build machine's paths (`/home/claude`, `/tmp`, `/mnt`).	Info	Documented in the build instructions
K8	Build 119 (the rollback) has the C7 problem: press **Resume saved game** the first time it is offered.	Info	In the rollback guide

Release readiness: Ready for live acceptance testing. Not marked ready for unrestricted live use until your real-device game passes (J). No known open defect would stop a game; K1 is a rule decision waiting for you.